

Restored films, limited reach

Film restoration, from Page 1

The academy recently signed a Memorandum of Understanding with NFAI for digitising and restoring 23 of the 68 analogues, according to Registrar Vinod Chandra.

The state government will maintain the archive which will hold both digitally restored and analogue copies.

According to Sunzu Bachaspatimayum, national award-winning filmmaker and secretary of the Manipur State Film Development Society (MS-FDS), at least 10 Manipuri films and five documentaries have been digitised but “the process has been very challenging”. “The efforts are being undertaken by Imphal-based SN Chand Cine Archive and Museum, and MSFDS with the help of National Film Archive of India, Pune, Film Heritage Foundation (FHF), Mumbai and others,” he said.

‘Brojendragee Luhongba’ (Brojendra’s Marriage, 1973), the first film produced by a Manipuri director, was the first movie from the northeastern states to be digitised and archived. The negative of the film, directed by S N Chand, was lost and the positive reel donated by producer-director Makhonmani Mongsaba was the only surviving source. Aribam



Pain is temporary, film is permanent. Our filmmakers toiled endlessly, endured much because they held that belief. We have to act now to uphold their principle. Great cinema has great value for a healthy society

Intiaz Ali
Filmmaker

Syam Sharma’s ‘Ishanou’ was restored in 2023 in collaboration with FHF.

“Two reels of the movie were so damaged that they could not be repaired,” he said. The digitally restored film was released in May 2022. “It was very hard and challenging. It took about one year and cost us Rs 7 lakh,” he said.

Assam has produced more than 1,000 movies since the first Assamese movie, ‘Joymoti’, was released in 1935. But only about 300 have been digitised and archived so far. The state has not been able to make much progress in restoring old movies, including those produced by legendary filmmaker Bhupen Hazarika. ‘Chameli Memsaab’ (1975), the Assamese classic directed by Abdul Majid, was the first film to be digitised.

“The main challenge in restoration is non-availability of the negative or the prints. Several classics by legendary Bhabendra Nath Saikia, Jahnu Barua and Sanjeeb Hazarika are also not available,” says film critic Nayan Prasad.

The Centre for International Film Research and Archives under the Kerala State Chalachitra Academy is carrying out the archiving. ‘Kavyamela’ (1965), ‘Olavum Theeravum’ (1969), ‘Yavanika’ (1982), ‘Vastuhara’ (1991), ‘Ormakal undayirikkanam’ (1995) and ‘Bhoothakkannadi’ (1997) are the films archived so far.

Analogue preservation

Film reels are essentially chemical substances and tropical climates can accelerate their deterioration. Both humidity and high temperature affect film. Nitrate films were no longer manufactured after 1955, when acetate became the commonly used film base.

High humidity can also cause acetate film to deteriorate. Cellulose acetate interacts with moisture, producing a strong, unpleasant odour, known as vinegar syndrome. This is the first stage of decomposition. A distinct smell is followed by the gradual appearance of a white, powdery substance on the film.

“It is, in many ways, like the human body. It will eventually deteriorate health-wise if we do not take care of it. Philosoph-

ically, we also have to accept that film will die, just like the human body,” Aparna notes.

Sometimes, it may not be possible for a national or state archive to take care of everything, given its limited resources and shortage of trained personnel. Restoration comes at the end of the process. The priority is to properly maintain, preserve and conserve the film reels themselves.

The process

Digitisation begins with the physical film reel. Before scanning, the film is inspected and cleaned, sometimes using ultrasonic cleaning, to remove dust and other contaminants. It is then scanned frame by frame. The process is deliberately slow because old film can be brittle or damaged, and tears can make it particularly difficult to handle. Newer scanners can work around some forms of damage, including sprocket tears, while sprocketless scanners can scan film even when its sprocket holes are no longer intact.

The result is a massive digital file. A feature film can generate anywhere between 12 and 18 TB of raw scan data. Once scanning is complete, the footage is reviewed in controlled, darkened rooms, allowing the film’s digitised condition to be examined in close detail. This is also where the distinction between digitisation and restoration becomes clear.

Not every film that is digitised necessarily needs to be restored. If the scan reveals only minor dust or speckles and the film remains watchable, basic fixes and colour grading may be enough. But if there are extensive scratches, other visual damage or problems with the soundtrack, the film may be sent for picture and audio restoration.

“Ideally, restoration begins only after digitisation. This also means restoration can be done in phases. If an archive has 100 films, it can digitise all 100, carrying out basic colour grading and fixes, and then prioritise a smaller number for restoration when funding becomes available,” Aparna shares.

Capital intensive

The process, however, does not end with restoration. Archiving is capital-intensive at every stage — from storing the original reels in climate-controlled vaults to scanning and restoring them, and then preserving the enormous amount of digital data generated in the process.

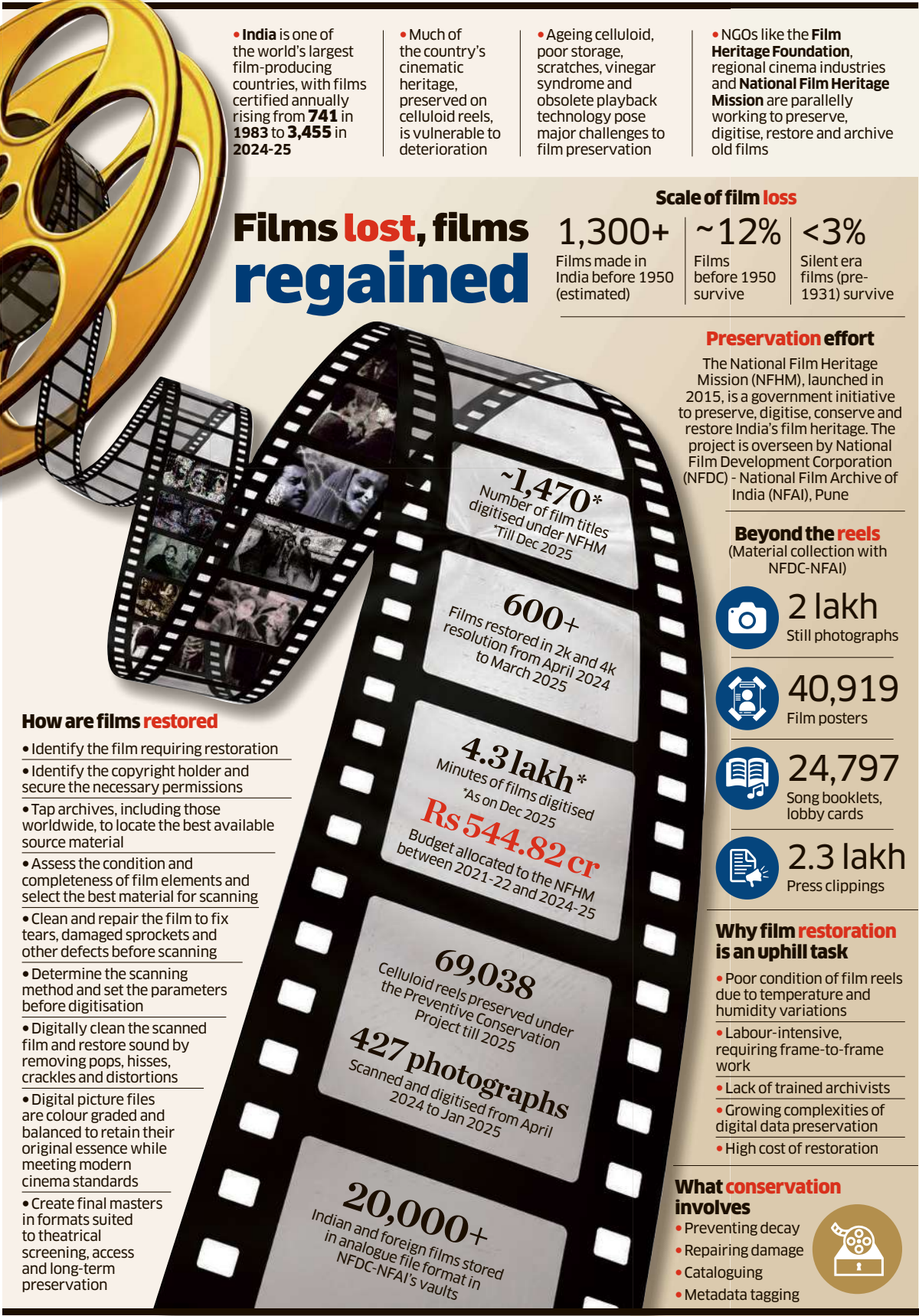
A restored film also has to be converted into different formats depending on how it will be accessed. A 4K restored Digital Cinema Package (DCP) for a feature film can be anywhere between 300 and 400 GB, requiring significant digital storage. Despite the growing reliance on cloud storage, large film files are expensive to store and difficult to transfer.

“Even with widespread internet access, downloading hundreds of gigabytes can be unreliable, with large transfers sometimes breaking midway. The challenge, therefore, is not simply to digitise films, but to ensure that the digital copies remain safely stored and accessible for years to come,” Aparna notes.

But digital restoration is not the end of the preservation process. Digital data is vulnerable to system failure, and files can be lost in a data crash, years down the line. If the original analogue material has not been preserved, there would be nothing left to rescan. The negative itself, therefore, remains the ultimate preservation object.

The 2K, 4K, 8K

With so many films being restored, how many actually become accessible to



1,325 films await digital preservation

Language	Number of films in NFDC-NFAI collection requiring digital preservation*
Hindi	607
Marathi	226
Bengali	91
Gujarati	82
Malayalam	77
Tamil	76
Punjabi	51
Bhojpuri	34
Telugu	29
Kannada	24
Others	28

*Data as of 2023



PIC CREDIT: FILM HERITAGE FOUNDATION

Cost of film digitisation and restoration at NFAI

Services	Per minute charges (In Rs)	Estimate charge of two hour film (In Rs)*
Film digitisation and grading in 4K	2,216	2.66 lakh
Film restoration along with digitisation 4K	22,836	27.4lakh

*Cost varies based on condition of original source material and complexity of restoration



Source: NFDC report, NFAI, MIB annual report, PIB, Indian Film Institute, media reports Compiled by DIVYASHRI MUDAKAVI DH Graphic: GURU PRASATH R

the public? While films such as ‘Umrao Jaan’ and Guru Dutt’s works have been released in theatres for brief periods, many restored films remain largely inaccessible.

Martin Scorsese’s The Film Foundation, which restores films from across the world, hosts monthly virtual screenings through its Restoration Screening Room,

with the films later made available on the Criterion Channel. But these platforms remain difficult for the average Indian viewer to access. Even Indian films restored in the country often remain out of public reach, unless they happen to be screened at a film festival.

Vidhu Vinod Chopra’s ‘1942: A Love Story’ was recently restored to 8K reso-

lution. In a statement, Chopra noted that the 8K version could not be screened in all Indian theatres due to the lack of adequate projection infrastructure.

Aparna believes the human eye cannot necessarily distinguish between 2K, 4K, and 8K — she considers these technical terms are used for marketing and sales. It is worth noting that only a handful of

Martin Scorsese's preservation project

PRANATI A S

US-based film director Martin Scorsese has been involved in film preservation for decades through The Film Foundation, which he founded in 1990. Its World Cinema Project (WCP), established in 2007, focuses on restoring films from regions whose cinematic heritage is at risk of being lost.

To date, 72 films from Africa, Asia, Eastern Europe, Central America, South America and West Asia have been restored, preserved and exhibited for audiences worldwide. The WCP also supports Restoration Film schools, intensive workshops that train students and professionals in film preservation and restoration. In 2017, it launched the African Film Heritage Project with the Pan African Federation of Filmmakers (FEPACI) and UNESCO, in collaboration with Cineteca di Bologna, to preserve African cinema.

In India, the WCP has worked with the Film Heritage Foundation (FHF), founded by filmmaker and archivist Shivendra Singh Dungarpur, on restorations of ‘Aranyer Din Ratri’ (1970), ‘Thamp’ (1978), ‘Kummatty’ (1979), and Ghatashraddha (1977). The films were screened at international film festivals.

premium or recently upgraded theatres in India currently support 4K projection, while the majority still operate on 2K.

Priority

Film critic and former artistic director of the Bengaluru International Film Festival N Vidyashankar believes that films considered classics and those of cultural significance should be prioritised for restoration.

NFAI prioritises films based on critical acclaim, commercial success, cultural and historical importance, and most importantly based on the overall deteriorating condition of the film reels, as per their Prioritization Framework for Digitization and Restoration of Films under NFHM.

Aparna believes films should also be viewed through the lens of time and space. “If I can see Pune, or any other city, as it was in the 1950s or 60s captured in a film, that becomes historical documentation, and that holds real value for a researcher, a film historian, or even a filmmaker researching for a period film,” she says.

The same is true of the Lucknow captured by Muzaffar Ali in ‘Umrao Jaan’ (1981), now restored in 4K; the Bombay streets and coffee houses of Guru Dutt’s ‘Pyaasa’ (1957); and the Bengal that lives on in Ritwik Ghatak’s films — its landscapes, rivers, railway tracks and lives shaped by Partition.

(With inputs from Sumir Karmakar in Guwahati, Arjun Raghunath in Thiruvananthapuram, ETB Sivapriyan in Chennai and Mrityunjay Bose in Mumbai)

Have your say

To express your opinion, e-mail us at: insight@deccanherald.co.in

Agniveers, from Page 1

Services seek to retain more tech-savvy Agniveers

“The government has not yet promulgated the terms and conditions of services of Agnivees in CAPF and other forces. Clarity on the terms and conditions is essential,” Lt Gen Rakesh Sharma (retd), a former Adjutant General of the Indian Army, told *DH*.

The Agnipath scheme was officially announced by the Union government on June 14, 2022 and came into effect in September 2022. The scheme was introduced to fundamentally restructure the recruitment of personnel below officer rank in the Indian Army, Navy and Air Force, with the objective of reducing the military’s growing salary and pension bill.

The scheme was proposed by the Indian Army on the “Tour of Duty” model, which would allow soldiers (Other Ranks) to be enlisted for one tour, a specific period, with pay and perks comparable to those of regular troops. They could be recruited for another tour if they agreed. The scheme was essentially the Other Ranks equivalent of the Short Service Commission for officers and was meant only for the Army.

Drastic modification

The Narendra Modi government drastically modified the proposal, making it the sole recruitment route for troops in

all three Services. The scheme was named Agnipath, with the soldiers, sailors and air personnel recruited under it called Agniveers. The implication is that these are young men and women who have been tested under the toughest of conditions and are ready to face all challenges in the service of the nation. From the original idea of retaining 75% of them after five years, it was decided that 75% would leave after four years with an exit package of Rs 11.7 lakh. There are a few other allowances as well.

The government hoped the scheme would bring down the ballooning salary and pension bill on the one hand, while lowering the average age of fighting troops on the other. After four years, it is not clear if any assessment has been made on either of these aspects. Such assessments, even if made, are not in the public domain.

The significance of the first batch goes beyond the number of personnel leaving or being retained. It also marks the beginning of the first large-scale transition of trained military personnel into the wider security and employment ecosystem. For the armed forces, it will be an opportunity to assess whether four years is sufficient to train, employ and develop personnel for

increasingly technology-intensive military roles, whereas for the government, it will be the first test of whether those leaving the services can successfully transition into other careers.

One of the stated objectives of the scheme was to lower the age profile of the fighting troops. But for the 25% of Agniveers who will be retained, the retirement age after serving for another 15 years will be 37 years, as against the current practice of 33 years. Does this mean that the scheme will require a tweak at a later date? The scheme was also found distasteful in rural India, where a job in the armed forces determines the social matrix. How will the lateral absorption in CAPF or Assam Rifles (where the issue of regular pensions is still sub judice) be viewed? These are questions to which there are no easy answers.

Evolutionary process

Former Army Chief Gen Upendra Dwivedi didn’t rule out a revision of the scheme. In an interview to *DH*, days before his retirement in June, he said, “The scheme remains an evolutionary process. The full cycle of the first batch is yet to be completed, and any final assessment must be based

on experience, operational feedback and long-term performance. If at a later stage, feedback from the formations and operational requirements indicate the need for refinement at a later stage, the matter can be examined institutionally.”

The three Services are understood to have approached the government with a request to enhance the percentage of Agniveers to be retained in their respective services. The Army and Air Force reportedly sought 50% retention while the Navy wanted to retain 75% of its trained sailors. The move comes a year after Operation Sindoor, during which Agniveers showed their capability. The Services now want to retain these tech-savvy Agniveers. The formation of dedicated tech-heavy units such as Bhairav and Baaz battalions may be another key factor compelling the forces to seek an increase in the retention of trained and operationally experienced workforce.

Last time a political demand was raised to review the scheme, Prime Minister Modi used the 25th anniversary of Kargil Vijay Diwas to assert that the Agnipath programme would continue as announced, without any change. It remains to be seen what he does this time.



Agniveer Vayu trainees during a combined passing out parade in Chennai. PTI